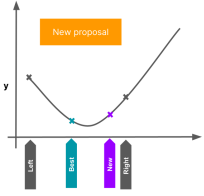
Optimization in Machine Learning

Univariate optimization
Golden ratio

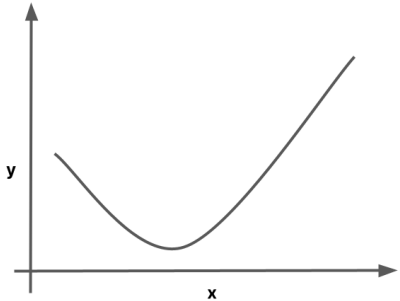


Learning goals

- Simple nesting procedure
- Golden ratio

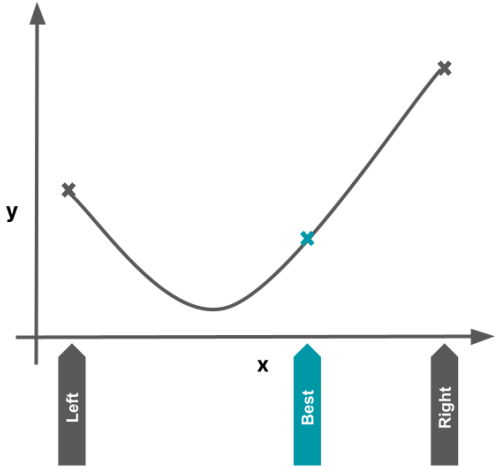
UNIVARIATE OPTIMIZATION

- Let $f : \mathbb{R} \to \mathbb{R}$
- **Goal**: Iteratively improve eval points
- Assume function is unimodal
- Will not rely on gradients, so this also works for black-box problems



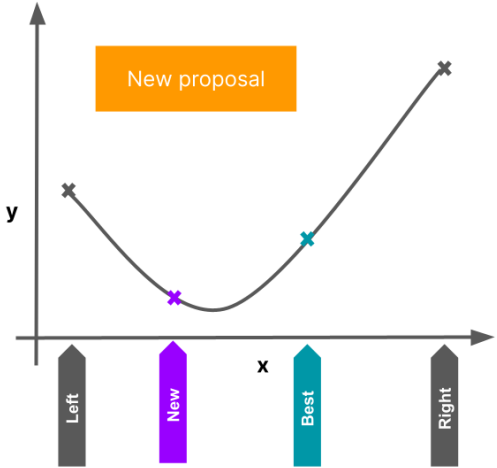
SIMPLE NESTING PROCEDURE

Always maintain three points: left, right, and current best.



SIMPLE NESTING PROCEDURE

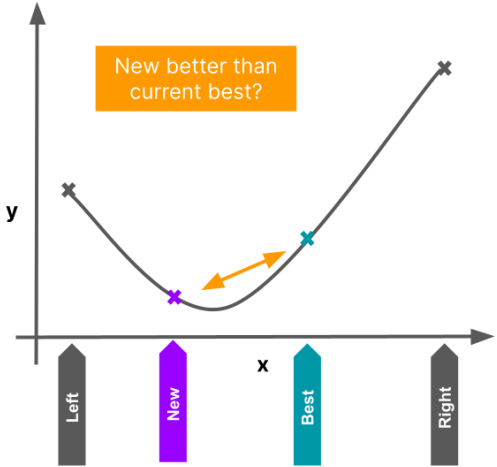
Propose random point in interval.

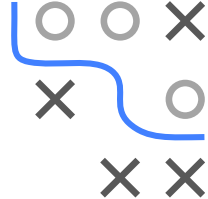


NB: Later we will define the optimal choice for a new proposal.

SIMPLE NESTING PROCEDURE

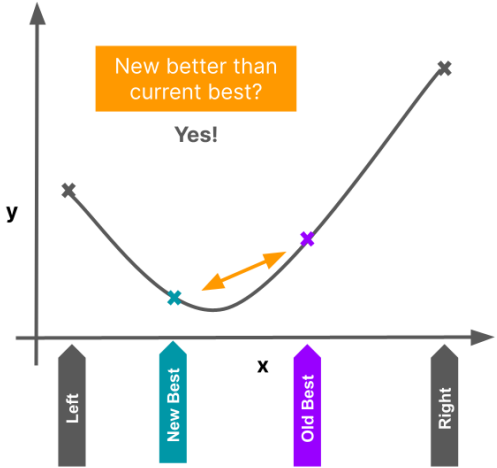
Compare proposal against current best.





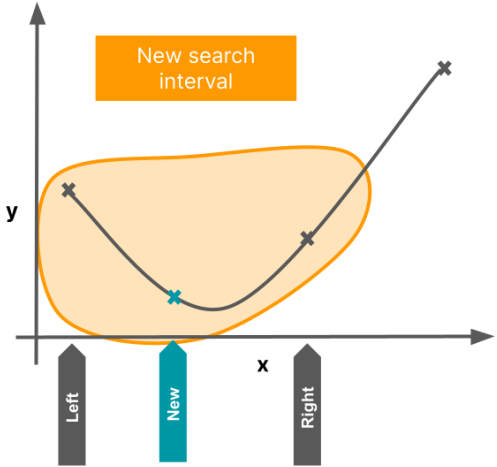
SIMPLE NESTING PROCEDURE

If it is better: proposal becomes current best.



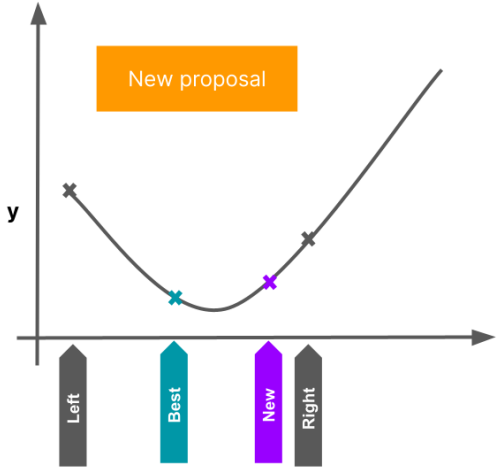
SIMPLE NESTING PROCEDURE

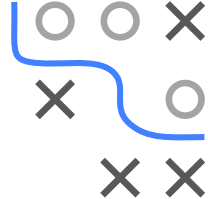
New search interval: around current best.



SIMPLE NESTING PROCEDURE

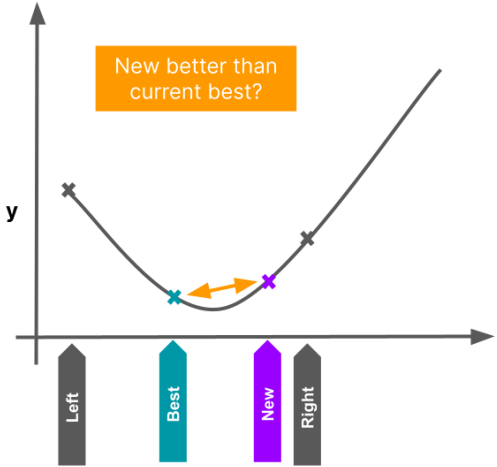
Propose a random point.

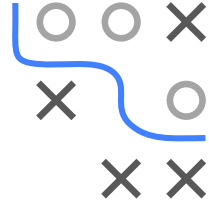




SIMPLE NESTING PROCEDURE

Compare proposal against current best.





SIMPLE NESTING PROCEDURE

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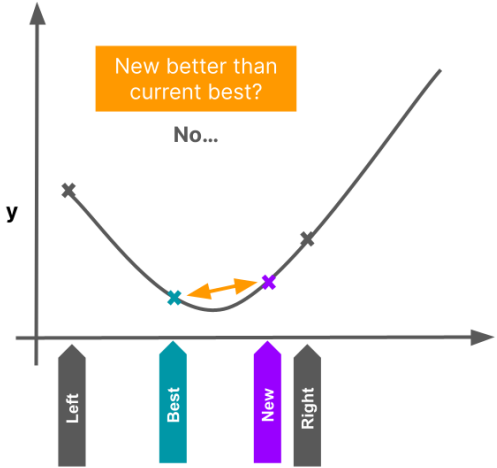
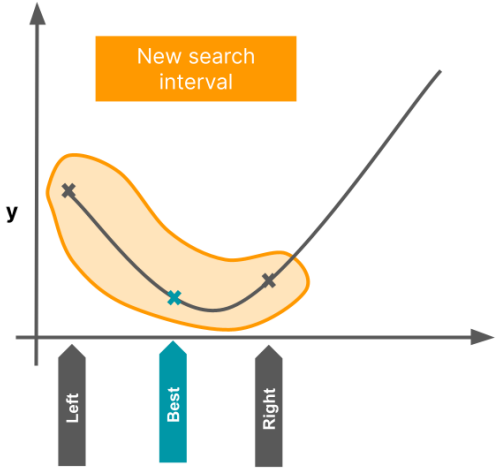


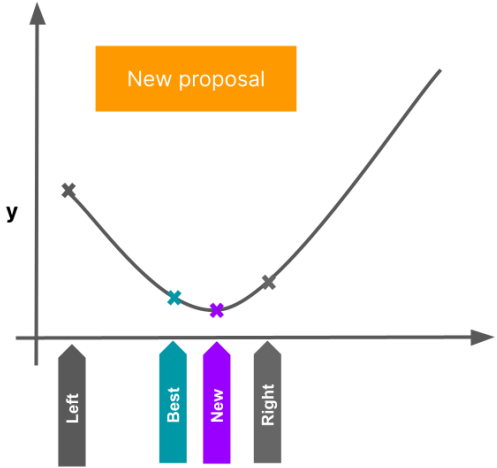
SIMPLE NESTING PROCEDURE

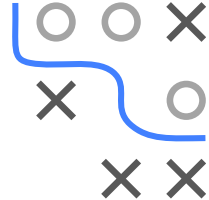
New search interval: around current best.



SIMPLE NESTING PROCEDURE

Propose a random point.





SIMPLE NESTING PROCEDURE

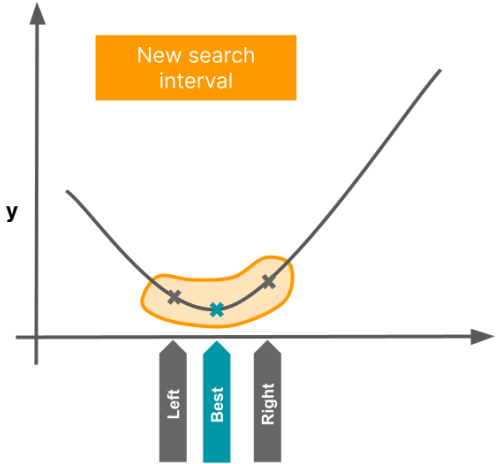
Compare proposal against current best.

SIMPLE NESTING PROCEDURE

If it is better: proposal becomes current best.

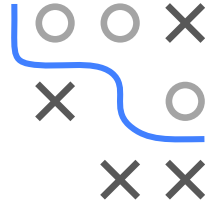
SIMPLE NESTING PROCEDURE

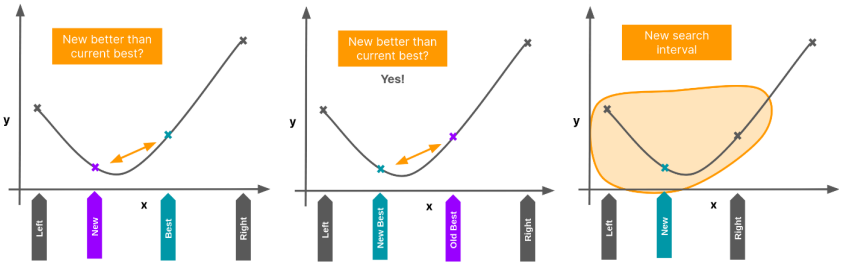
New search interval: around current best.



SIMPLE NESTING PROCEDURE

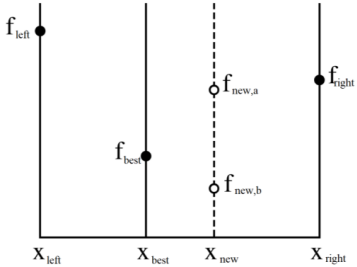
- **Initialization**: Search interval $(x^{\text{left}}, x^{\text{right}})$
- $x^{\text{left}} < x^{\text{right}}$
- Choose x^{best} randomly
- For $t = 0, 1, 2, \ldots$
- Choose x^{new} randomly in $[x^{\text{left}}, x^{\text{right}}]$
- If $f(x^{\text{new}}) < f(x^{\text{best}})$: $x^{\text{best}} \leftarrow x^{\text{new}}$
- New interval: Points around x^{best}





GOLDEN RATIO

- **Key question:** How can x^{new} be chosen better than randomly?
- **Insight 1:** Always in bigger subinterval to maximize reduction
- **Insight 2:** x^{new} symmetrically to x^{best} for uniform reduction

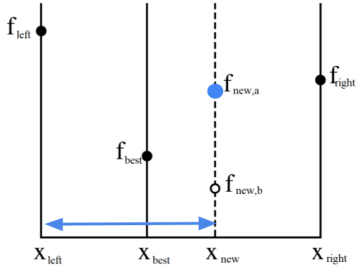


- Consider two hypothetical outcomes x^{new}: $f_{new,a}$ and $f_{new,b}$

GOLDEN RATIO

- If $f_{new,a}$ is the outcome, x_{best} stays best
- We search around x_{best}:
-

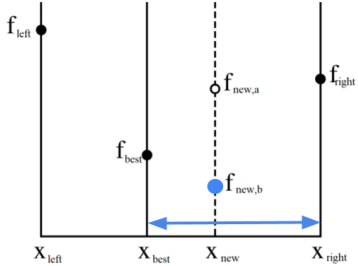
$$[x_{left}, x_{new}]$$



GOLDEN RATIO

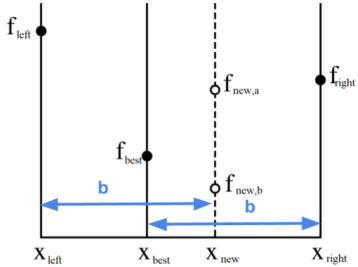
- If $f_{new,b}$ is outcome, x_{new} becomes best point
- We search around x_{new}:
-

$$[x_{best}, x_{right}]$$



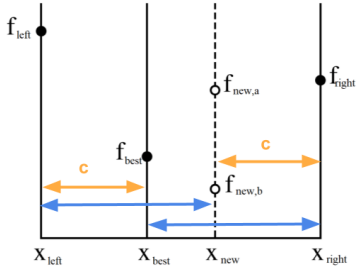
GOLDEN RATIO

- For uniform reduction, require the two potential intervals equal sized
- $b := x_{right} - x_{best} = x_{new} - x_{left}$



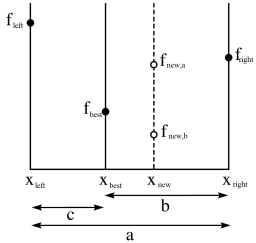
GOLDEN RATIO

- One iteration ahead: require again the intervals to be of same size
- $c := x_{best} - x_{left} = x_{right} - x_{new}$

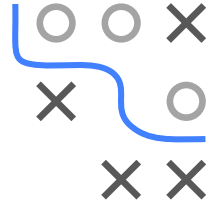


GOLDEN RATIO

- To summarize, we require:
- $a = x^{right} - x^{left}$
- $b = x_{right} - x_{best} = x_{new} - x_{left}$
- $c = x_{best} - x_{left} = x_{right} - x_{new}$

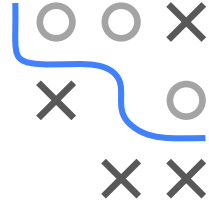


GOLDEN RATIO



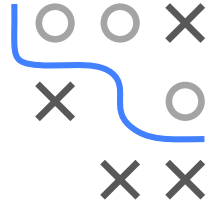
- We require the same percentage improvement in each iteration
- For φ reduction factor of interval sizes
- a to b, and b to c
- $$\varphi := \frac{b}{a} = \frac{c}{b}$$
- $$\varphi^2 = \frac{b}{a} \cdot \frac{c}{b} = \frac{c}{a}$$
- Divide $a = b + c$ by a:
- $\frac{a}{a} = \frac{b}{a} + \frac{c}{a}$
- $1 = \varphi + \varphi^2$
- $0 = \varphi^2 + \varphi - 1$
- Unique positive solution is $\varphi = \frac{\sqrt{5}-1}{2} \approx 0.618$

GOLDEN RATIO



- With x^{new} we always go φ percentage points into the interval
- Given x^{left} and x^{right} it follows
- $x^{best} = x^{right} - \varphi(x^{right} - x^{left}) = x^{left} + (1 - \varphi)(x^{right} - x^{left})$
- and due to symmetry
- $x^{new} = x^{left} + \varphi(x^{right} - x^{left}) = x^{right} - (1 - \varphi)(x^{right} - x^{left})$

GOLDEN RATIO



- Termination criterion:
- A reasonable choice is the absolute error
- I.e. the width of the last interval:
- $|x^{best} - x^{new}| < \tau$
- In practice, more complicated termination criteria are usually applied
- For example in *Numerical Recipes in C, 2017*:
- $|x^{right} - x^{left}| \leq \tau(|x^{best}| + |x^{new}|)$
- is proposed as a termination criterion